EXAMPLE 4.5

A straight conducting wire is dropped horizontally from a certain height with its length along east – west direction. Will an emf be induced in it? Justify your answer.

Solution

Yes! An emf will be induced in the wire because it moves perpendicular to the horizontal component of Earth's magnetic field and hence it cuts the magnetic lines of Earth's magnetic filed.

4.1.4 Lenz's law

A German physicist Heinrich Lenz performed a series of experiments on electromagnetic induction and deduced a law to determine the direction of the induced current. This law is known as Lenz's law.

Lenz's law states that the direction of the induced current is such that it always opposes the cause responsible for its production.

Faraday discovered that whenever magnetic flux linked with a coil changes, an electric current is induced in the circuit. Here the flux change is the cause while the induction of current is the effect. Lenz's law says that the effect always opposes the cause. Therefore, the induced current would flow in a direction so that it could oppose the flux change.

By incorporating Lenz's law into Faraday's law, the equation (4.2) is rewritten as

$$\varepsilon = -\frac{d(N\Phi_B)}{dt} \qquad (4.3)$$

The negative sign signifies that the direction of induced emf is such that it opposes the change in magnetic flux.

To understand Lenz's law, let us consider two illustrations in which we find the direction of the induced current in a circuit.

Illustration 1

Consider a uniform magnetic field, with its field lines perpendicular to the plane of the paper and pointing inwards. These field lines are represented by crosses (×) as shown in Figure 4.6(a). A rectangular metallic frame *ABCD* is placed in this magnetic field,

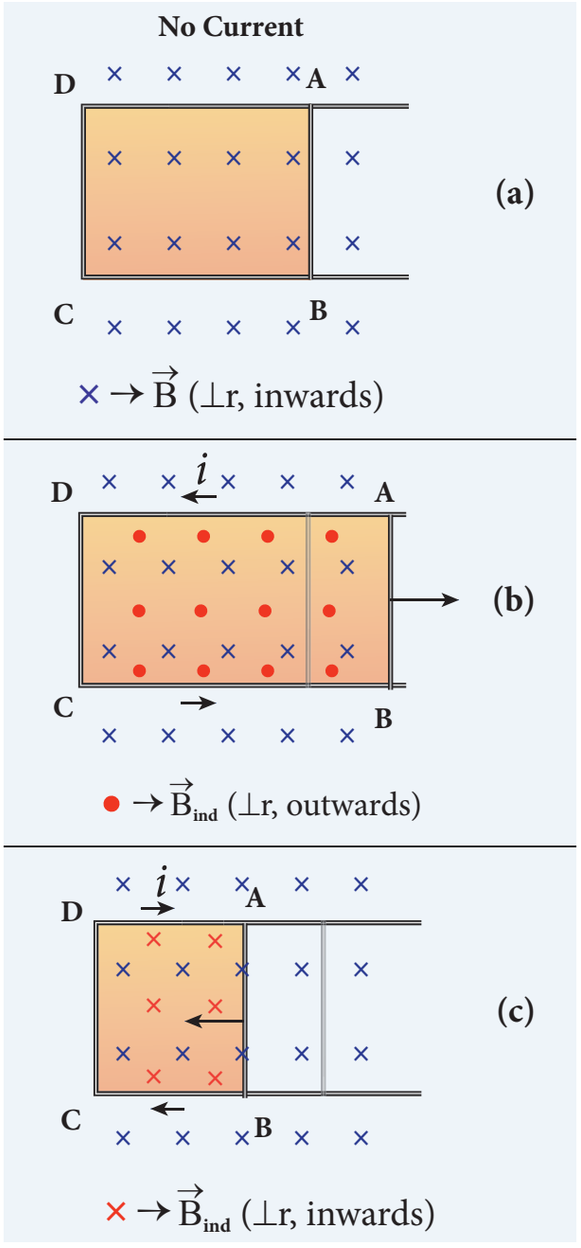


Figure 4.6 First illustration of Lenz's law

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